Yunfan Chen

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Research Interests: Climate Variability; Tropical Climate Dynamics; Ocean–Atmosphere Interactions; Paleoclimate Proxies; Coral Paleoclimatology; Climate Modeling

Education

University of Michigan, Ann Arbor, Michigan, U.S.A.

- Ph.D. Candidate in Earth and Environmental Sciences (GPA: 4.00/4.00) 08/2023 – 05/2027
- Graduate AI and Data Science Certificate Program 01/2026 – 05/2027
- Museum Science Communication Fellows Certificate 2025

South China Sea Institute of Oceanology, Chinese Academy of Sciences, Guangdong, China

- M.S. in Physical Oceanography (GPA: 3.61/4.00) 09/2020 – 06/2023

Hohai University, Jiangsu, China

- B.S. in Marine Science (WES Calculator GPA: 3.82/4.00) 09/2016 – 06/2020

Publications

- **Yunfan Chen**, Julia Cole, Samantha Stevenson, Midhun Madhavan, Yan Du, Yingying Feng, Joelle Tanuputri, Kelsey Dyez, 2026. Diagnosing CESM2 Biases in Northwest Indian Ocean Warming and Its Implications for India Summer Monsoon Precipitation. **Geophysical Research Letters**, 53(17): e2026GL123060. DOI: 10.1029/2026GL123060.
- Joelle Tanuputri, Julia Cole, Kelsey Dyez, Samantha Stevenson, Flora Luo, **Yunfan Chen**, Yingying Feng, 2026. Coral Sr/Ca from Venezuela Reconstruct Large-Scale Variability in Atlantic Sea Surface Temperature and Regional Hydroclimate Extremes. **Paleoceanography and Paleoclimatology**, 41(2): 2025PA005379. DOI: 10.1029/2025PA005379.
- **Yunfan Chen**, Yan Du, Zesheng Chen, 2023. Indo-Western Pacific Ocean Capacitor Events Recorded by Coral Proxies in the South China Sea. **Palaeogeography, Palaeoclimatology, Palaeoecology**, 609: 111315. DOI: 10.1016/j.palaeo.2022.111315.
- Yan Du, Fan Wang, Tianyu Wang, Weiwei Liu, Linlin Liang, Ying Zhang, **Yunfan Chen**, Jiaxing Liu, Wei Wu, Kefu Yu, Jing Zhang, 2023. Multi-Scale Ocean Dynamical Processes in the Indo-Pacific Convergence Zone and Their Climatic and Ecological Effects. **Earth-Science Reviews**, 237: 104313. DOI: 10.1016/j.earscirev.2023.104313.
- **Yunfan Chen**, Tianran Chen, Shangmin Long, Zesheng Chen, Yan Du, 2022. Reconstruction of Climate Variability of the Tropical Indian Ocean Based on the Coral $\delta^{18}O$ Data from Four Representative Sites. **Journal of Tropical Oceanography**, 41(01): 82-93. DOI: 10.11978/2021011.

Manuscripts under review

- **Yunfan Chen**, Julia Cole, Samantha Stevenson, Yingying Feng, Heidi Barnett, Joelle Tanuputri, Kelsey Dyez, Tim McClanahan. Indo-Pacific Interactions in the Industrial Era: Insights from Coral Records. Fourth-round minor revision submitted, **Journal of Climate**.
- Yan Xia, Yong-Fu Lin, Jin-Yi Yu, **Yunfan Chen**, Mike Pritchard. Asymmetric Atlantic–Pacific Interactions on Decadal Timescales Underrepresented in Climate Models. In revision, **Journal of Geophysical Research: Atmospheres**.
- Joelle Tanuputri, Julia Cole, Samantha Stevenson, Kelsey Dyez, Jonathan Overpeck, **Yunfan Chen**, Yingying Feng.

New 254-year Coral-based Reconstruction of Tropical North Atlantic Temperature Supports External Forcing Pacing Multidecadal Variability. Submitted to **Nature Communications**.

Manuscripts in preparation

- **Yunfan Chen,** Julia Cole, Yong-Fu Lin, Kelsey Dyez, Jin-Yi Yu, Yan Du, Samantha Stevenson, Joelle Tanuputri. A 387-year Coral Record Reveals Intermittent Solar Modulation of Boreal Wintertime Indian Ocean SST. In preparation for **Nature Communications**.
- Yingying Feng, Samantha Stevenson, Se-Yong Song, Julia Cole, Aoyun Xue, Na Wan, **Yunfan Chen**, Joelle Tanuputri. Ocean Basin Interactions Drive El Niño Initiation Following Tropical Volcanic Eruptions. In preparation for **Nature Geoscience**.
- Yingying Feng, Samantha Stevenson, Julia Cole, Aoyun Xue, **Yunfan Chen**, Joelle Tanuputri. Changes in ENSO Teleconnections to the Tropical North Atlantic and Tropical Indian Ocean Over the Last Millennium. In preparation for **Journal of Climate**.
- Zhangzhe Zhao, Agus Santoso, Guojian Wang, Wenju Cai, **Yunfan Chen**. Long-Term Salinity Changes in the Tropical Indian Ocean and Their Impacts on Climate.

Selected Conference Presentations

- **Yunfan Chen**, Julia Cole, Samantha Stevenson, Yingying Feng, Joelle Tanuputri, Kelsey Dyez. Assessing the Role of Northwest Indian Ocean Warming in Long-term Northeast India Drying. Poster, *OSM*, Glasgow, Scotland 02/2026
- **Yunfan Chen**, Julia Cole, Samantha Stevenson, Yingying Feng, Joelle Tanuputri, Kelsey Dyez. Examining the Role of the Pacific Ocean in the Unusual Warming of the Western Indian Ocean Over the Past Century. Poster, *PAGES YSM and OSM*, Shanghai, China 05/2025
- **Yunfan Chen**, Julia Cole, Samantha Stevenson, Yingying Feng, Joelle Tanuputri, Kelsey Dyez. How Has the Tropical Indian and Pacific Ocean Interaction Changed Over the Industrial Era? Insights From Coral Records. Poster, *AGU Fall Meeting*, Washington, DC, U.S.A. 12/2024
- **Yunfan Chen**, Julia Cole, Kelsey Dyez. Coral-Based Reconstruction of Climate Variability in the Western Indian Ocean. Poster, *MGU*, Ann Arbor, Michigan, U.S.A. 04/2024

Patents and Inventions

- **Yunfan Chen**. 2019. A Device for Growing Marine Coral. ZL201821918130.3, filed 11/21/2018, and issued 08/30/2019. Link
- **Yunfan Chen**. 2018. A Self-cleaning Fish Tank. ZL201510887166.4, filed 12/07/2015, and issued 08/07/2018. Link
- **Yunfan Chen**. 2016. A Circulating Oxygenation System for Fish Tank. ZL201521012586.X, filed 12/09/2015, and issued 04/20/2016. Link
- **Yunfan Chen**. 2016. A Self-cleaning Fish Tank. ZL201521001141.1, filed 12/07/2015, and issued 04/20/2016. Link

Research Experience

- **Indo-Pacific Climate Variability, Interactions, and Regional Impacts** 08/2023 – Present

Research Assistant, Paleoclimate and Common Era Lab (Advisor: Prof. Julia Cole) Michigan, U.S.A.

(1) Multi-Proxy Paleoclimate Reconstruction in the Western Indian Ocean

- Reconstructed sea surface temperature in the western Indian Ocean using multi-proxy methods (composite plus scale, principal component analysis, principal component regression, and regularized expectation maximization).
- Investigated Indo-Pacific climate variability and interactions using CESM1 Last Millennium Ensemble simulations.
- Manuscript resubmitted following fourth-round minor revision at **Journal of Climate**.

(2) Model Diagnosis of Long-Term Indian Ocean Warming and Its Monsoon Impacts

- Investigated ocean–atmosphere coupling mechanisms linking Pacific variability to Indian Ocean warming using CESM2 Large Ensemble and Pacific Pacemaker experiments.
- Diagnosed model–data mismatches in Indian Ocean warming and associated monsoon impacts over the past century.
- Published in **Geophysical Research Letters** (Link).

(3) Decadal–Multidecadal Variability in the Indian Ocean Modulated by Solar Forcing

- Generated coral $\delta^{18}O/\delta^{13}C$ from the long-lived coral *Diploastrea heliopora* collected at Mafia Archipelago, Tanzania.
- Reconstructed 387-year sea surface temperature variability in the Indian Ocean.
- Investigated atmospheric and oceanic mechanisms linking solar forcing to Indian Ocean variability using ERA5 reanalysis and solar-forcing experiments from CESM1 and GISS.
- Manuscript in preparation.

• **Climate Variability and Climate Dynamics of the Tropical Oceans** 09/2020 – 06/2023

Research Assistant, State Key Laboratory of Tropical Oceanography (Advisor: Prof. Yan Du) Guangdong, China

(1) Coral-Based Reconstruction of Tropical Indian Ocean Climate Variability

- Analyzed coral $\delta^{18}O$ records from four tropical Indian Ocean sites to reconstruct regional climate variability during 1880–1999.
- Published in **Journal of Tropical Oceanography** (Link).

(2) Indo-Pacific Ocean Interactions Recorded by Coral Proxies in the South China Sea

- Investigated Indo–western Pacific Ocean capacitor events and their influence on the South China Sea using coral $\delta^{18}O$ and Sr/Ca.
- Published in **Palaeogeography, Palaeoclimatology, Palaeoecology** (Link).

(3) Ocean Dynamical Processes in the Indo-Pacific Convergence Zone (IPCZ)

- Synthesized multi-scale ocean dynamical and environmental processes in the Indo-Pacific Convergence Zone and their climatic and ecological impacts.
- Published in **Earth-Science Reviews** (Link).

• **Numerical Simulation of Atlantic Meridional Overturning Circulation** 04/2021 – 06/2021

Graduate Researcher, University of Chinese Academy of Sciences (Advisor: Prof. Tianjun Zhou) Beijing, China

- Analyzed AMOC responses to freshwater forcing and wind stress using sensitivity experiments with the LICOM3.0 ocean model.

• **Deriving Three-Dimensional Temperature–Salinity Structure from Satellite Observations** 03/2018 – 06/2019

Project Technician, Hohai University (Advisor: Prof. Xidong Wang) Jiangsu, China

- Reconstructed three-dimensional subsurface temperature and salinity fields from satellite observations using an inversion method with vertical-gradient constraints.

Teaching Experience

Graduate Student Instructor, Department of Earth and Environmental Sciences, University of Michigan, Ann Arbor

• **EARTH 223 / ENVIRON 233** Introductory Oceanography Lab 01/2025 – 04/2025; 09/2025 – 12/2025

- Led weekly introductory oceanography laboratory sections, explaining quantitative Earth science concepts to undergraduate students with diverse technical backgrounds.
- Guided students through data analysis, scientific interpretation, and collaborative problem-solving.

Laboratory and Fieldwork Experience

Laboratory Experience:

- Generated coral $\delta^{18}O/\delta^{13}C$ from *Porites* spp. and *Diploastrea heliopora*. 08/2023 – Present
- Instruments: Thermo Delta V Plus isotope ratio mass spectrometer; scanning electron microscope (SEM).

Field Experience:

- *Member*, Summer Camp for College Students "Blue Ocean Initiatives" at CAS, Guangdong, China 07/2019
- *Member*, Marine Comprehensive Survey Practice & Jiangsu Offshore Marine Environment Survey 05/2019
  - Conducted hydrographic measurements including sea surface temperature, salinity, current velocity, and tidal levels in the Yancheng coastal region.
  - Instruments: Secchi disk; water colorimeter; acoustic Doppler current profiler (ADCP); conductivity–temperature–depth profiler (CTD); optical backscatter sensor (OBS); direct-reading current meter.
- *Leader*, Argo Center, Second Institute of Oceanography, Ministry of Natural Resources, Zhejiang, China 10/2018
  - Led a student team in training on oceanographic observations and data processing.

Fellowships and Funding

- Rackham Predoctoral Fellowship 05/2026 – 05/2027
- NSF-supported Graduate Student Research Assistant 01/2026 – 04/2026; 01/2024 – 12/2024
- Rackham International Student Fellowship 05/2025 – 08/2025
- Earth and Environmental Sciences NOVAK Fellowship 09/2023 – 12/2023

Honors and Awards

- China National Scholarship (top 0.2%) 2022
- Student Award of Merit, Chinese Academy of Sciences (top 10%) 2020 – 2022
- Academic Merit Scholarship, Hohai University (top 5%) 2016 – 2020
- Outstanding Student Award, Hohai University (top 2%) 2017 – 2020
- Technology and Innovation Scholarship, Hohai University (top 2%) 2017 – 2019
- Jiangsu Provincial Government Scholarship for Overseas Studies (top 2%), King's College London 2018

Skills

- **Languages:** Mandarin (Native); English (Professional Working Proficiency); Japanese (Basic Proficiency)
- **Programming & Software:** Python; MATLAB; Fortran; Linux/Unix; High-Performance Computing (HPC; Great Lakes); GitHub; Globus; Adobe Illustrator; Photoshop; Premiere Pro
- **Climate & Ocean Modeling:** CESM; GISS-E2; LICOM; ROMS; SWAN
- **Data Analysis:** Regression; time-series analysis; PCA/EOF; principal component regression; machine learning; spectral and wavelet analysis; climate model evaluation; heat-budget analysis; RegEM
- **Professional Certification:** Google Advanced Data Analytics Professional Certificate